

Simple Tester for Dual Operational Amplifiers

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Here is a simple audible tester for dual operational amplifiers (op-amps). Dual op-amps such as TL061, TL072, TL082, NE5532, RC4558 and MC1458 are widely used in hobby circuits and professional equipment. Before soldering them on a printed circuit board (PCB), it is a good idea to test them for general functionality. This is especially obligatory when op-amps from old projects or old equipment are being reused.

Circuit and working

Circuit diagram of the tester is shown in Fig. 1. It comprises only basic components such as resistors, capacitors and connectors in addition to the dual op-amp that needs to be tested.

The circuit produces two different sound signals when both the op-amps are functioning well. The circuit actually has two square-wave generators with bipolar power supply. It operates in the range of $\pm 4.5\text{V}$ to $\pm 15\text{V}$. The power supply is applied to connector

CON1 as shown in Fig. 1.

The frequency of generated sound depends on capacitors C1 and C2, and is in the range of around 450 Hz to 4.5 kHz.

Most popular operational amplifiers produce square wave, without any problem, in this frequency range. Usually C1 and C2 are in the range of 1nF to 10nF, but their values are not critical. However, values of C1 and C2 should be different from each other to ensure that both the op-amps generate different audible signals.

The signals can be heard through stereo headphones having resistance of 32 ohms or more. The headphones are connected to connector JACK1. Ordinary low-cost stereo headphones of an MP3 player or a Walkman may be used here. Resistors R9 and R10 can be

reduced to 510 ohms if the signal from the headphones is not strong enough, but usually that is not needed.

The circuit starts working immediately after completion and does not need any adjustment. Working of the circuit is simple. Suppose you want to test a dual op-amp such as TL072. Insert the IC in the socket and power on the circuit. If you hear two different audio frequencies through the headphones it means both the op-amps are fine.

Construction and testing

An actual-size, single-side PCB for the tester is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable box. Fix connectors CON1 and CON2 for power supply and stereo headphones on front side of the box.

Solder IC base on the PCB for inserting the dual op-amps to be tested.

Two 9V batteries are enough for the circuit. For some op-amps even two 4.5V battery packs will do.

To check the circuit for proper functioning, verify +5V supply at TP1 and -5V at TP2, with respect to TP0. The generated audio signals can be verified at TP3 and TP4. ●

Test Points

Test point	Details
TP0	0V, GND
TP1	+5V
TP2	-5V
TP3	Around 450 Hz
TP4	Around 4.5 kHz

PARTS LIST

Semiconductors:

IC1 - TL072 dual operational amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R8 - 100-kilo-ohm
R9, R10 - 1-kilo-ohm

Capacitors:

C1 - 0.01 μF ceramic disk
C2 - 0.001 μF ceramic disk
C3, C5 - 0.33 μF ceramic disk
C4, C6 - 100 μF , 16V electrolytic

Miscellaneous:

CON1 - 3-pin connector
CON2 - Female audio jack
- 8-pin IC base
- 32-ohm headphones

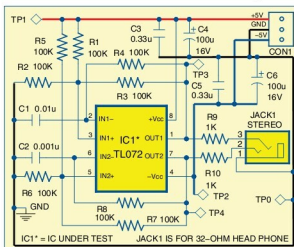


Fig. 1: Circuit diagram of the simple tester for dual operational amplifiers

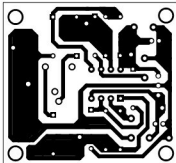


Fig. 2: An actual-size, single-side PCB for the tester

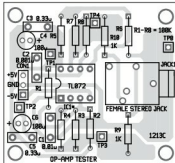


Fig. 3: Component layout for the PCB

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